Algorithms Analysis and Design

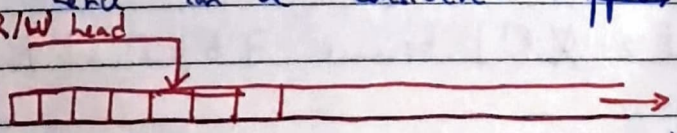
Algorithm:- No singular answer, operational definitions exist, like "Set of steps to reach a goal"

10th Hilbertian Question:- Give an algo to solve Random Diophantine Eqn. [1900]

The definition of an algorithm has answered multiple questions like "What is a machine?", "What is a computer?", etc.

Axioms

1) Information travels at finite speeds. [Spl. Theory of Relativity]
   → Random Access Memory is a myth. All memory is sequential, hence can be considered ~~tapes~~ as tapes.



Speed of the R/W head is 1 cell/per unit time.

2) Finite volume in space can be used to store or retrieve only finite amounts of information reliably [Quantum Mechanics]
   → This implies that we can not store ∞ precision of an irrational numbers. Thus using (perfectly precise) pi or the like in an algorithm is not allowed.
   Thus each cell can only store a finite amount of data [Based on A1 & A2]

   Data in any cell $\in \Gamma$: Finite set of tape alphabet

3) Only finitely many instructions can be packed a priori. [ISA is finite]
   → Computers are finite; hence.

Algorithms book by Das Gupta

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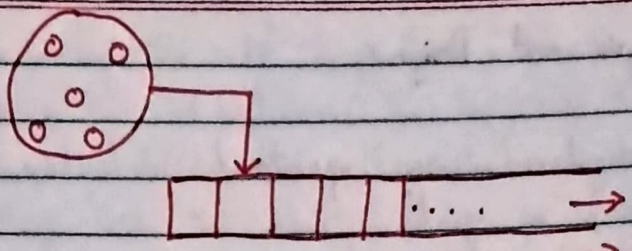


→ Anything that

(set of steps/processes etc)

Church Turing Thesis

Algorithm :- Any algorithm that can be simulated using a Turing Machine

Turing Machine :- A Turing Machine is a 7-tuple

$\langle Q, \Gamma, \delta, q_{start}, q_{accept}, q_{reject}, \Sigma \rangle$

Q → Finite ~~state~~ set of states

Γ → Finite set of state alphabets (that can fill cells)

$\delta : \Gamma \times Q \rightarrow \Gamma \times Q \times \{L, R\}$

$\Sigma \subset \Gamma$ $[\Sigma \not\subset \Gamma]$ because $\exists b \in \Gamma$ s.t. $b \notin \Sigma$

A Turing Machine can have another Turing Machine as its input

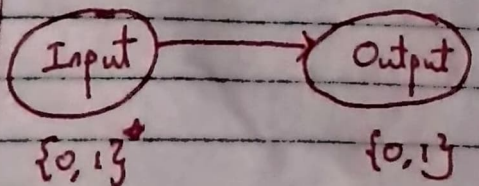
$U_{TM}(\langle M_{TM} \rangle, x) = M_{TM}(x)$

Where U_{TM} & M_{TM} are Turing Machines x is the input to M and $\langle\rangle$ is the string typecast.

U_{TM} is the Universal Turing Machine

What is a computational problem?

Decision Problems



Test

Quick brown Fox Jumps over the lazy Dog

Problems having multiple bits can be considered as multiple problems having single bit outputs

All ~~problems~~ computational problems are thus ~~considered~~ defined as problems that ~~are~~ have a set of inputs which can be ~~bipartitioned~~ bi-partitioned into a T/F inputs. Since the two partitions are MECE, just being given the input set and one of its subsets is sufficient. All computational problems are hence decision problems.

Decision Problems = (Membership Queries in) Languages.

A Language $L \subseteq \{0,1\}^*$

Given a graph G, is it connected or not? Decision problem

(or)

Does G belong to the set of all connected graphs Membership Query

$G \in \{G' \mid G' \text{ is a connected graph}\}$

Is a given number n prime?

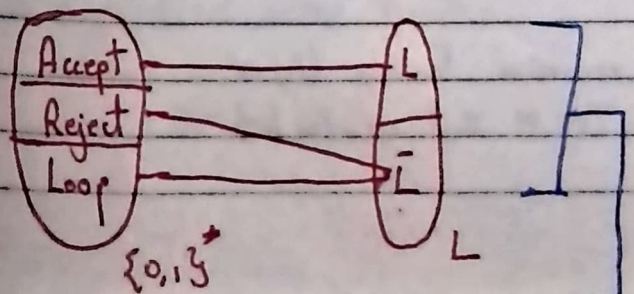
↓

$n \in \{i \mid i : \text{is prime}\}$

Problems are languages, solutions are Turing Machines

Turing Machines have three states, Accept, Reject and Loop

Languages have only two states, L & $\bar{L}$



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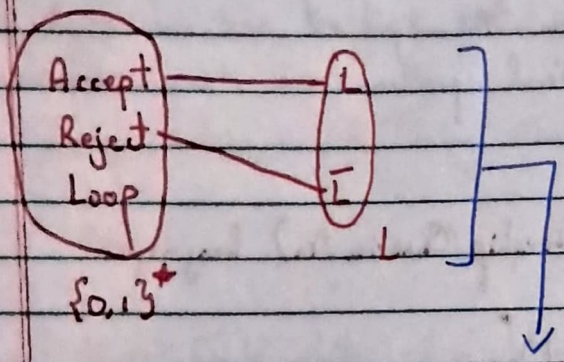
Definition:- A language L is said to be recognized by a Turing Machine M if:-

$\forall x \in L$; M(x) accepts.

else ; M(x) rejects.

Hence the language L of M $[L(M)] = \{x \mid M(x) \text{ accepts}\}$



Definition:- A language L is said to be ~~recognize~~ decided by TM M is

$\forall x \in L$; M(x) accepts

else ; M(x) rejects

A TM that decides L also recognizes L, but vice versa does not hold.

Theorem:- There are languages that are unrecognizable

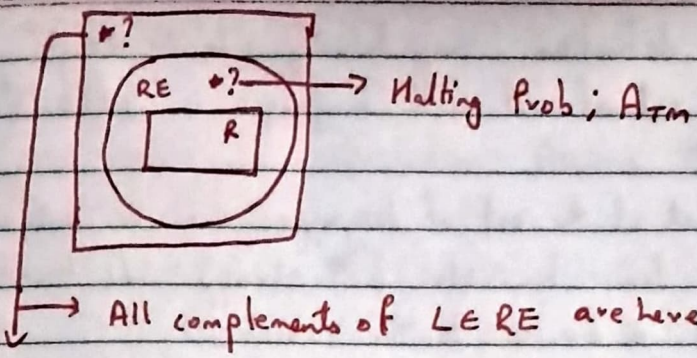
Theorem:- There are languages that are recognizable but undecidable

Defn:- The class of recursively ~~enumerated~~ enumerable (RE) languages is

$RE = \{L \mid \exists$ a TM M that recognizes $L\}$

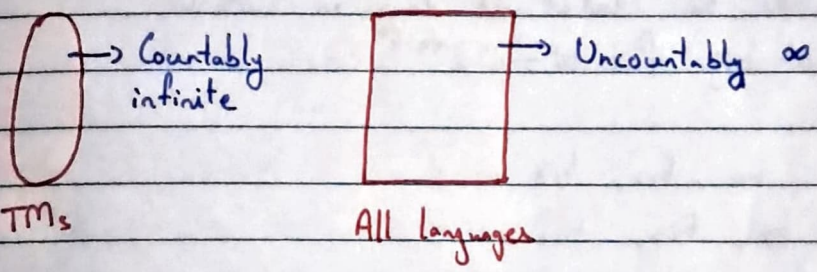
Defn:- The class of recursive languages (R) is

$R = \{L \mid \exists$ a TM M that decides $L\}$



How do you show that ~~$\exists$~~ $\exists$ languages here?

Take the set of all TMs and the set of all problems



- Any TM can be stored in a machine with finite bit strings, hence the no. of TMs $\leq$ no. of finite len bin strings
Thus TMs $\subseteq \{0,1\}^*$ → They can be enumerated
How?

Theorem:- $\{0,1\}^*$ is countable

Proof:- $f: N \to \{0,1\}^*$ must be a bijection
Consider the following enumeration of $\{0,1\}^*$
Shorter first, lexicographic ordering for eq lens.

$\epsilon, 0, 1, 00, 01, 10, 11, \underbrace{000, \ldots}_{8}, \underbrace{0000 \ldots}_{16}, \ldots$

This is a bijective map cause every string has a fixed index.
[Find a closed order formula]

Theorem:- The set of languages is uncountable.

Proof:- By definition, any language $L \subseteq \{0,1\}^*$.
So the set of all such languages is $P(\{0,1\}^*)$

Each subset of the set of languages can be uniquely identified by an ∞ len bin str [i^{th} str[i] = 0/1 based on whether the i^{th} element $\in$ or $\notin$ to the subset]

Any $L \equiv$ inf len bin str.

Assume that the set of all langs is countable.
Then $\exists$ a bij $f : N \to 2^{\{0,1\}^*}$

$f(1) = b_{11}, b_{12}, b_{13} \ldots$
$f(2) = b_{21}, b_{22}, b_{23} \ldots$
$f(3) = b_{31}, b_{32}, b_{33} \ldots$

Since its countable, all N have been mapped to these $f(i)$
Now we diagonalization to create a new language
$L = \overline{b_{11}}, \overline{b_{22}}, \overline{b_{33}}, \overline{b_{44}}, \ldots$
This L has no mapping and is distinct from all $f(i)$ in at least 1 pos. Hence its uncountably ∞.

The class RE are those programs which give yes, no or loop, but never returns a wrong answer.

Given a TM M and input x, will M accept x?

$A_{TM} = \{\langle M, x\rangle \mid M \text{ accepts } x\}$

This problem is recognizable, not decidable $\longrightarrow \in RE$

<M> → Program typecast to a str to be used as input!

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Theorem:- A_{TM} is undecidable.

Proof:- We prove by contradiction. Assume that A_{TM} was decidable
→ ∃ a TM H that decides A_{TM}.
→ ∀ $\langle M, x \rangle \in L$, $H(\langle M, x \rangle)$ accepts;
∀ $\langle M, x \rangle \notin L$, $H(\langle M, x \rangle)$ rejects; [not loop]

Consider a TM D which does the following
i) On input M
ⓐ Run ~~H (<M>, m)~~ H (M, <m>)
ⓑ Return ¬ H's result (accept on reject and vice versa)

Execute D (<D>)
On input <D>, Run H (D, <D>)
If H accepts, then rejects → (D, <D>) ∈ A_{TM}
This implies D accepts <D> then D rejects ::
Contradiction

If H rejects, D accepts → (D, <D>) ∉ A_{TM}
This implies D rejects <D> then D accepts.
Contradiction

(or)

	$\langle M_1 \rangle, \langle M_2 \rangle \cdots \langle M_i \rangle$
M_1	$H(M_1, \langle M_2 \rangle)$
M_2	
⋮	
M_i	

Use diagonalization create a new D, but $H(D, \langle D \rangle)$ is a contradiction.
- - - That's how diagonalization works!

Theorem:- If $L \in RE$ and $\bar{L} \in RE$, then $L \in R$

Proof:- ~~$L \in RE \Rightarrow \forall x \in L, \exists$ TM M s.t M accepts x~~

$\forall$

i) $L \in RE \Rightarrow \exists$ a TM M s.t
$\forall x \in L$; M accepts x
$\forall x \in \bar{L}$; M rejects / loops on x

ii) $\bar{L} \in RE \Rightarrow \exists$ a TM M' s.t
$\forall x \in \bar{L}$; $\bar{M}$ accepts x
$\forall x \in L$; $\bar{M}$ rejects / loops on x

Create a decider ~~the~~ by running M and M' in parallel.
M accepts $x \in L$ and M' accepts $x \in \bar{L}$.

Theorem:- ~~$A_{TM} \notin RE$~~ $\bar{A}_{TM} \notin RE$

Proof:- $A_{TM} \in RE$; $A_{TM} \notin R$
By contradiction and prev theorem, $\bar{A}_{TM} \notin RE$.

$TIME(n^k)$ → Problems solved in $O(n^k)$, where $k \in$ Rational
$TIME(n^{k+\epsilon})$ → Superset of $TIME(n^k)$

Reduction:- A language L_1 is mapping reduced to L_2 if $\exists$ a computable fn $f: \{0,1\}^* \to \{0,1\}^*$ s.t. $\forall x \in \{0,1\}^*$, $x \in L_1 \rightleftharpoons f(x) \in L_2$

↳ $L_1 \leq_m L_2$

~~**Defn:-** Language $A \leq_m B$~~

$f(\langle M, x \rangle) = \{ H(M,x) \wedge [M(x)] \}$

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Examples:-

Let $HALT_{TM} = \{ \langle M, x \rangle \mid M \text{ halts on input } x \}$

Lemma:- $A_{TM} \leq_m HALT_{TM}$

Proof:- Decider for A_{TM} (using the one for $HALT_{TM}$)

Let H decide $HALT_{TM}$

On input $\langle M, x \rangle$:-

→ Run $H(M,x)$

→ If H rejects, REJECT

→ If H accepts, ACCEPT or REJECT based off $M(x)$

Theorem:- If $A \leq_m B$ and $B \in R \rightarrow A \in R$

Theorem:- If $A \leq_m B$ and $A \notin R \rightarrow B \notin R$

Completeness Theory:-

Suppose $\forall A \in RE$, $A \leq_m HALT_{TM}$.

Then $HALT_{TM}$ is RE-complete

Theorem:- $HALT_{TM}$ is RE-complete

↳ Anything in RE

Proof:- Let TM M recognize A & H decides $HALT_{TM}$

Construct a decider for A.

On input x :

Run $H(M,x)$. If

$EQUAL_{TM} = \{\langle M_1, M_2\rangle \mid L(M_1) = L(M_2)\}$

Theorem:- $EQUAL_{TM}$ is unrecognizable

Proof:- $\overline{A_{TM}}$ is unrecognizable ($\overline{A_{TM}} \notin RE$)

Let E recognize $EQUAL_{TM}$

ω

M_1 : On input [redacted], Reject

M_2 : On input ω, if $x \neq \omega$ Reject
If M accepts x, ACCEPT.
Else Reject

If $E(M_1, M_2)$ accepts $\rightleftharpoons$ M does not accept x.

$(M, x) \in A_{TM}$

Divide and Conquer.

Mergesort, Binary Search, Median of Medians.

Fourier Transforms:-
Input:- Two polynomials A(x) & B(x) of degree n-1
Output:- Their product $C(x) = A(x) \cdot B(x)$

$A[n] \rightarrow$ coefficients of A
$B[n] \rightarrow$ coefficients of B
$C[2n-1] \rightarrow$ coefficients of C

$$A(x) = \sum_{i=0}^{n-1} a_i x^i$$

$$B(x) = \sum_{i=0}^{n-1} b_i x^i$$

$$C(x) = \sum_{i=0}^{2n-1} c_i x^i$$

$$c_i = \sum_{j=0}^{i} a_j b_{i-j} \longmapsto$$ But this is an $O(n^2)$ algorithm

But Fast Fourier Transforms can do it in $O(n \log n)$

Algorithm :-

$O(n)$ ← i) ~~Evaluate~~ Select $2n-1$ or more points $x_1, \ldots x_m$ $[m \geq 2n-1]$

ii) Evaluate $A(x_i)$ and $B(x_i)$ $\longmapsto O(n^2)$ [n-pts with n-1 substitutions per pt]

iii) Multiply $A(x_i) \times B(x_i) \Rightarrow C(x_i)$ $\longmapsto O(n)$

iv) Interpolate $C(x_i)$'s to obtain $C(x)$ $\longmapsto O(n^2)$

Let $A(x)$ be the polynomial

$$A(x) = A_e(x^2) + x A_o(x^2)$$

$A(x) = 5x^4 - 3x^3 + 7x^2 + x + 1$

$A_e(x) = 5x^2 + 7x + 1$

$A_o(x) = -3x + 2$

For any $A(x_i) = A_e(x_i^2) + x_i A_o(x_i^2)$

$A(-x_i) = A_e(x_i^2) - x_i A_o(x_i^2)$

$T(n) = 2T(n/2) + O(n)$

But ~~$A_e(x_i^2) = A_{ee}(x_i^4) + x_i^2 A_{eo}(x_i^4)$~~

~~$A_e(-x_i^2)$~~

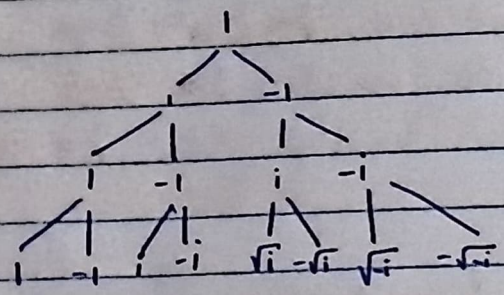
→ +/− space

A: $\{x_1 \quad -x_1 \quad x_2 \quad -x_2 \quad \cdots \quad x_d \quad -x_d$

$\{ \quad x_1^2 \quad x_2^2 \quad x_d^2$

→ Not +/− space ⇒ Can't recurse further.

Hence, we pick complex nos to evaluate.
Let begin from the bottom.



4^{th} roots of unity

8^{th} roots of unity

m^{th} roots of unity

$e^{i\frac{2\pi}{m}j}$ m is an exact power of $2 \geq 2n-1$

$j \rightarrow [1, m]$

Goal: To evaluate a polynomial A on the m^{th} roots of unity
$1, \omega, \omega^2, \cdots, \omega^{m-1}$

Input: A[]
Output: $[A(1), A(\omega), \cdots, A(\omega^{m-1})]$

$$A = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_d \end{bmatrix} \qquad A(x) = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^d \\ 1 & x_2 & x_2^2 & \cdots & x_2^d \\ & & \vdots & & \\ 1 & x_{d+1} & x_{d+1}^2 & \cdots & x_{d+1}^d \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_d \end{bmatrix}$$

To evaluate our output, we multiply A with

$$\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^d \\ \vdots & & & & \\ 1 & \omega^{m-1} & \omega^{(m-1)\cdot 2} & \cdots & \omega^{(m-1)^2} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{m-1} \end{bmatrix} = \begin{bmatrix} A(1) \\ A(\omega) \\ \vdots \\ A(\omega^{m-1}) \end{bmatrix}$$

↳ i,j^{th} entry $\rightarrow \omega^{ij}$ $\forall\ i,j \in [0, m-1]$

↳ Discrete Fourier Transform.

~~FFT~~ FFT(A, ω):

```
if ω == 1:
    return A(1)
else:
    A_e = [A[0], A[2], ....]
    A_o = [A[1], A[3], ....]
    ~~FFT(A_e~~
    E = FFT(A_e, ω^2)
    O = FFT(A_o, ω^2)
    ~~for i in range(m-1):~~
        ~~A[i] = E[i/2] + ω^2 O[i/2]~~
        # E and O are half the size of A, hence we
        # must recd at half the index
    for i in range((m•)/2):
        A[i] = E[i] + ω^i • O[i]
        A[m/2 + i] = E[i] - ω^i • O[i]
        # • Check penultimate page for the formula
```

Now, the second step in our algorithm (check antipenultimate page) runs in $O(n \log n)$

In step 1, we pick $m \geq 2n-1$ where m is a power of 2, which are just the m^{th} roots of unity.

I'm lost uwu [We've fucked]

Interpolation

$$\begin{bmatrix} M \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{m-1} \end{bmatrix} = \begin{bmatrix} c(1) \\ c(\omega) \\ \vdots \\ c(\omega^{m-1}) \end{bmatrix}$$

$$\downarrow$$

$$M_{ij} = \omega^{ij}$$

$$\begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{m-1} \end{bmatrix} = M^{-1} \begin{bmatrix} c(1) \\ c(\omega) \\ \vdots \\ c(\omega^{m-1}) \end{bmatrix}$$

$$M^{-1} = \frac{1}{m} M(\omega^{-1}) \quad [Y_{um}\ Y_{um}]$$

or

$$[Y_{um}]^*$$

$$[\omega^{ij}]_{m \times m} \times [\omega^{-ij}]_{m \times m}$$

$$= \begin{bmatrix} m & & & \\ & m & & \\ & & m & \\ & & & \ddots \end{bmatrix}$$

This (somehow! some fucking how!) makes the last step $O(\log n)$

Karatsuba's Algorithm:-

Large integer multiplication [Two n-bit +ve integers]

Normal multiplication takes $O(n^2)$

Fastest way to multiply large integers is still FFT by converting the no. into a polynomial $f(x)$ where $f(2) = int$.

Sorting Detour:-

Assuming all numbers are bounded.

Countsort :- Store them in an array at loc x_i
↓
count(x_i)

Radix sort:- Apply countsort on every position, units, tens, etc. until the largest place.

End detour:-

Multiplying 2 complex numbers $(a+ib)(x+iy)$

$\Rightarrow ax - by + i(bx + ay)$

Looks like it needs 4 real products, but one can just use 3 :-

(i) $p_1 = (x_1 + y_1)(x_2 + y_2)$

(ii) $p_2 = x_1 \cdot x_2$

(iii) $p_3 = y_1 \cdot y_2$

Similarly, for n bit numbers a & b

$a = a_L \cdot 2^{n/2} + a_R$

$b = b_L \cdot 2^{n/2} + b_R$

$ab = a_L b_L 2^n + 2^{n/2}(a_L b_R + a_R b_L) + a_R b_R$

We can use this to establish a recurrence, but it would be $O(n^2)$. Instead we apply Gauss' Logic, reducing it to 3 multiplications:-

$$P_1 = (a_L + a_R)(b_L + b_R)$$
$$P_2 = a_L b_L$$
$$P_3 = a_R b_R$$

and $ab = P_2 2^n + 2^{n/2}[P_1 - P_2 - P_3] + P_3$

Recurrence on this results in

$$T(n) = 3T(n/2) + O(n)$$
$$= O(n^{\log \frac{3}{2}})$$

$$T(n) = 3T(n/2) + O(n)$$
$$= 3[3T(n/4) + O(n/2)] + O(n)$$
$$= 3^2(T(n/2^2)) + 3n/2 + n$$
$$= \ldots$$
$$= 3^3 T(n/2^3) + 3^2 n/2^2 + 3^1 n/2^1 + n$$
$$= 3^i T(n/2^i) + n \sum_{j=0}^{i-1} 3^j/2^j$$

Let $i = \log_2 n$

$$T(n) = 3^{\log_2 n} + n \sum_{j=0}^{\log n - 1} 1.5^j$$

$$\cong O(n^{\log \frac{3}{2}})$$

Strassen's Algorithm:-

$$[A]_{n \times n} \quad [B]_{n \times n} \rightarrow \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]_{n \times n} \quad \left[\begin{array}{c|c} E & F \\ \hline G & H \end{array}\right]_{n \times n}$$

$$= \begin{bmatrix} AE+BG & AF+BH \\ CE+DG & CF+DH \end{bmatrix}_{n \times n}$$

$T(n) = 8\,T(n/2) + \underbrace{O(n^2)}_{\text{Addition}}$

$= 8\left[8\,T(n/4) + O\left((n/2)^2\right)\right] + O(n^2)$

$= \ldots\ldots$

$\cong O(n^3)$

BAD! :<

Strassen's ~~technique~~ technique :- He does it in 7 mults

$T(n) = 7(T(n/2)) + O(n^2)$

$= \ldots$

$= O(n^{\log_2 7})$

The product would then be

$$\left[\begin{array}{c|c} P_5 + P_6 + P_4 - P_2 & P_1 + P_2 \\ \hline P_3 + P_4 & P_5 - P_7 + P_1 - P_3 \end{array}\right]$$

$P_1 = A\,(F-H)$

$P_2 = (A+B)\,H$

$P_3 = (C+D)\,E$

$P_4 = D\,(G-E)$

$P_5 = (A+D)\,(E+H)$

$P_6 = (B-D)\,(G+H)$

$P_7 = (A-C)\,(E+F)$

Fibonacci?

$F_n = F_{n-1} + F_{n-2}$

$F_0 = 0$

$F_1 = 1$

Recursion → 2^n → bad :<

Iteration → n → okay

↳ but for large n, add becomes $O(n)$ — we have $O(n)$ additions.

↳ n^2 for large n

$$\begin{pmatrix} F_1 \\ F_2 \end{pmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{pmatrix} F_0 \\ F_1 \end{pmatrix}$$

$$\begin{pmatrix} F_2 \\ F_3 \end{pmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}^2 \begin{pmatrix} F_0 \\ F_1 \end{pmatrix}$$

$$\vdots$$

$$\begin{pmatrix} F_n \\ F_{n+1} \end{pmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}^n \begin{pmatrix} F_0 \\ F_1 \end{pmatrix}$$

↳ $\log n$ multiplications

Greedy Algorithms

Used in optimization problems

Ex :- Shortest path - single source, Huffman encoding.

Optimum Substructure Property:- ~~Sup~~ Subparts of a problem must also be using the most efficient solution.

If $S \to V$ is the shortest path, then $S \to T \Rightarrow S \to U + U \to T$,

In such cases we usually recurse, given the first step to be taken.

Greedy Choice Property :- Picking the right first step.

We need both Optimum Substructure and Greedy Choice to have an efficient Greedy Algo

If only the Optimum Substructure holds, we try out all possibilities, but this is slow, unless we have :-

Subproblems

Overlapping ~~Substructure~~ Property: If we are able to create a DAG with the subproblems. We toposort first and solve subproblems in toporder

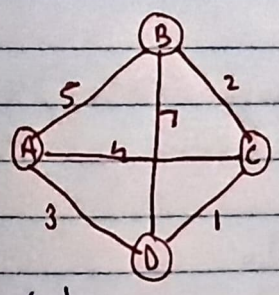
Greedy is a spl case of DP where the DAG is a path.

Minimum Spanning Trees:-

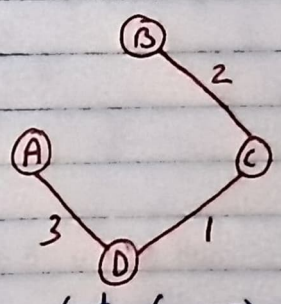
Input:- Edge Weighted Simple Undirected Path.

Subgraph of a tree on all its vertices

Cost of tree = Σ edge wts.



Cost :- 21



Cost = 6 → MST

Output:- Least cost Spanning Tree T of G

The Cut Property:-

Let $G = (V, E)$

Suppose $X \subset E$ belongs to some MST of G.
Then let V be divided into S and V/S such that no edge spans from S → V/S or vice versa.

Then the least cost edge e b/w S & V/S is s.t $X \cup \{e\}$ is part of some MST of G.

Algo:- [Kruskal]

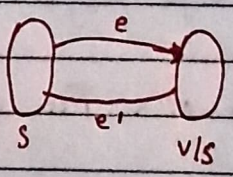
Sort edges by wt.
Add edges in that order ~~until~~ unless cycle is formed
the MST

Proof:- Suppose $X \cup \{e\}$ is part of T if X is a part of T.

$e \notin E(T)$

$e' \in T \rightarrow e' \in cut$

$wt(e) \leq wt(e')$



$(T \cup \{e\}) \setminus \{e'\}$ has to be connected cause $T \cup \{e\}$ formed a cycle.
↳ had n edges, so the graph $(T \cup \{e\}) \setminus \{e'\}$ has n-1 edges and is connected → is a tree T'

$cost(T') = cost(T) + wt(e) - wt(e')$.
$\leq cost(T)$

Since T is an MST, $cost(T') = cost(T)$
Hence T' is an MST

Prim's Algorithm:-

Let $X = \phi$

Repeat:

(i) Find a cut that doesn't contain X

(ii) Update $X = X \cup \{e\}$ where e is the min-cost edge in that cut.

Until $|V|-1$ edges are added.

Kruskals is preferred because of advanced data structures like union-find :- Find → $O(\log n)$, Union → $O(1)$ and tot complexity of $O(|E| \log |V|)$ without path compression.

With path compression ⇒ Find → $O(\log^* n)$

Activity Selection / Interval Scheduling Problems:

Input = $\{[s_1, f_1), [s_2, f_2), \cdots, [s_n, f_n)\}$

Non over lapping interval:

Output:- λ Maximum sized subset of compatible activities

Hint:- Activity A_i is part of some optimum soln.
Remove an activities compatible with A_i

Greedy Choice:- The activity with the least finish time is always a part of some optimum soln.

Algo:- Sort all to finish time. Initialize the first activity. Pick next activity that is compatible and continue

Knapsack Problem:-

Input:- Collection of items $\{(c_1, w_1), (c_2, w_2), \cdots, (c_n, w_n)\}$

Output:- Subset of items s.t total wt $\leq W$, maximize cost

Matroid Problem:-

Definition:- A matroid is a pair $\langle S, I \rangle$ where:

(a) S is a finite non-empty set

(b) I is a family of collection of subsets of S s.t. $[\phi \in I]$ (set of independent sets ↓)

(i) Hereditary Property:- If $A \subseteq S$ belongs to I, then $\forall B \subseteq A$, $B \in I$

(ii) Exchange Property:- If $A, B \in I$ and $|B| > |A|$, then $\exists x \in B \setminus A$ s.t $A \cup \{x\} \in I$

Question:- Given a Matroid $\langle S, I \rangle$ and given non-ve int wts to elements of S, find max wt element of I (or subset of S)

$w(A) = \sum_{x \in A} w(x)$. $x \in S$ & $w(x)$ = wt of x.

Ans:- Universal Greedy Algo for weighted matroids.

→ Sort the elements of S in non-increasing order of wts
→ $A \leftarrow \phi$
→ In that order, let x be the next element
if $A \cup \{x\} \in I$ then $A \leftarrow A \cup \{x\}$
Finally output A.

Proof of correctness / optimality

(i) Greedy Choice :- The elemet $x \in S$ with max wt belongs to some optimum solution.

Proof :- Let $A \in I$ be an optimum soln.

if $x \in A$:

pass

else: $[x \notin A \& \{x\} \in I]$

Let $A = \{a_1, a_2, \ldots, a_k\}$

From exchange property, we know that $\exists$ a $B = (A \cup \{x\}) \setminus \{a_i\}$ & $B \in I$

$\omega(B) = \omega(A) + \omega(x) - \omega(a_i) \geq \omega(A)$

a_i has ~~better~~ lesser wt than x cause x appeared before a_i in sorted order

$M = \langle S, I \rangle$ contracts to $M' = \langle S', I' \rangle$, $A = A' \cup \{x\}$

$S' = S - \{x\}$; I' : sets of I with x erased

$I' = \{ B \mid B \cup \{x\} \in I \}$

$\omega(A) = \omega(A') + \omega(x)$

Given an edge weighted graph $G = (V, E)$

Let $S_G = E$

$I_G = \{ E' \subseteq E \mid E' \text{ does not have a cycle} \}$

Theorem :- $M_G = \langle S_G, I_G \rangle$ is a matroid.

I_G is hereditary, cause if $A \in I_G$, all $s \subseteq A$ also $\in I_G$.

I_4 has the exchange property.

$|E_1| > |E_2|$ → forest with $|V| - |E_2|$ trees

↓

forest with exactly $|V| - |E_1|$ trees

E_1 has an edge (u, v) s.t u, v are in diff trees in E_2.
Add this edge to E_2, which will not form a cyclic graph.

$E_2 \in T \rightarrow E_2 \cup \{(u,v)\} \in I_4$

Hence M_4 is a matroid.

I_4 has the exchange property.

$|E_1| > |E_2|$ → forest with $|V| - |E_2|$ trees

↓

forest with exactly $|V| - |E_1|$ trees

E_1 has an l edge (u, v) s.t u, v are in diff trees in E_2.
Add this edge to E_2, which will not form a cyclic graph.

$E_2 \in I \rightarrow E_2 \cup \{(u,v)\} \in I_4$

Hence M_4 is a matroid.

Dynamic Programming :-

A bag of subproblems s.t it forms a DAG of smallest to largest

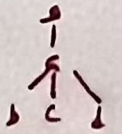
Shortest Path in a DAG.

Input : Weighted Directed Acyclic Graph with source s and dest D

Output :- Shortest path from s to D

Algo :- Topo sort the DAG

$P(s, D) = \min(P(s, x) + dP(x, D))$ ∀ x in paths b/w $s - D$
& $P(s, s) = 0$.
Recurse



Longest Path ?

Algorithm :- Toposort the DAG

$P(s,i) = \max (P(s,u) + d(u,i)) \; \forall \; u$

$P(s,s) = 0$ & return

Both run in ~~$O(E)$~~ $O(V+E)$

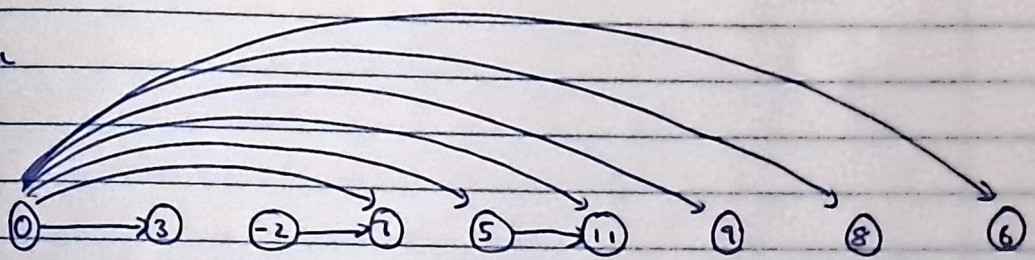
Longest increasing subsequence [LIS]

Input :- Array of nos

Output :- LIS → Strictly increasing

Subsequence → any no. can be erased, but order is fixed.

Ex



Similarly connect all nodes only to those nodes strictly greater than itself

Now find the longest path in the DAG

$L(i,j) = \max_{(u,j) \in E} (1 + L(i,u))$

Find the ~~longest~~ LIS for all nodes, but the nodes appearing

Complexity - $O(k \cdot |E|)$ → k is no. of source nodes
↳ [adjacency list]

in any prob LIS cannot be the starting node for another LIS because of greedy choice and optimum substructure

Edit Distance

min no. of edit operations (insert, delete, overwrite) to convert one str to another.

Ex:- EARTH & HEART

(i) insert H to the first
(ii) delete H from the last

SUNNY & SNOWY

(i) Overwrite U → N
(ii) " N → O
(iii) " N → W

Q) Given a currency $D = \{d_1, d_2, \ldots, d_n\}$ and a target V find the minimum set of coins.

Algo:- Init an empty sol set
Sort D in desc
Find the largest $d_i \in D$, $d_i \leq V$
sol set = sol set $\cup \{d_i\}$
$V \leftarrow V - d_i$
repeat.

Optimality:-
Assume our algorithm is not optimal.
Say our algorithm differs with the optimal soln at some position j. But at position j, our $q_j > o_j$ because of greedy choice. In every case, ~~our optimal~~ replacing a coin

is our greedy sol. results in an optimal sol O where $|G| \leq |O|$.
Hence, the greedy sol. is most optimal.

Q) Interval Scheduling.

Init empty array.
Sort input array by finish time asc & in case of ties by start time.
Pick first element and append to array.
Iterate and pick next interval with $s_i \geq f_0$

Repeat till done

Proof:-
Base Case :- When $A = \emptyset$, all jobs in A are compatible
Induction Hypothesis :- All jobs in $i < j$ in A are compat. bl
Induction Step :- ~~We add $i = j$~~ if j in A, & if all jobs $i < j+1$ in A are compatible : else all jobs in $i < j+1$ in A are compatible

Edit Distance :-

$X = x_1\ x_2 \cdots x_m$
$Y = y_1\ y_2 \cdots y_n$

Add as many blanks as necessary to make both of same length and such that the strings are aligned
Cost of alignment = no. of mismatches

Subproblems :-
(prefix) i^{st} i chars of X
(prefix) i^{st} j chars of Y

We call Edit(i,j) To try and reach the goal of Edit(r,m)

(a) x_i → $E(i,j) = E(i-1,j) + 1$

(b) y_j → $E(i,j) = E(i,j-1) + 1$

(c) x_i, y_j → $E(i,j) = \begin{cases} E(i-1,j-1) ; x_i = y_j \\ E(i-1,j-1) + 1 ; \text{else} \end{cases}$

$$Edit(i,j) = \min \begin{pmatrix} E(i-1,j) + 1 \\ E(i,j-1) + 1 \\ E(i-1,j-1) + diff(x_i, y_j) \end{pmatrix}$$

$Edit(0,j) = j$

$Edit(i,0) = i$

→ Base Case

	0	x_1	x_2	x_3	...	x_n
y_1						
y_2						
y_3						
.						
.						
y_m						$E(m,n)$

Fill row or column wise (because for any cell, the cells to the left, above and diagonally up and left must be filled)

		H	E	A	R	T
	0	1	2	3	4	5
E	1	1	1	2	3	4
A	2	2	2	1	2	3
R	3	3	3	2	1	2
T	4	4	4	3	2	1
H	5	4	5	4	3	**2** → E(n,m)

Blue Arrows show one possible way of deriving cell values, row wise

Q) Edit Distance b/w ~~Polyno~~ POLYNOMIAL & EXPONENTIAL

		P	O	L	Y	N	O	M
	0	1	2	3	4	5	6	7
E	1	1	2	3	4	5	6	7
X	2	2	2	3	4	5	6	7
P	3	2	3	3	4	5	6	7
O	4	3	2	3	4	5	5	6
N	5	4	3	3	4	4	5	6
E	6	5	4	4	4	5	5	6
N	7	6	5	5	5	4	5	6
T	8	7	6	6	6	5	5	6

Chain Matrix Multiplication :-

Input :- Chain of matrices

$$M_{P_0 \times P_1} \times M_{P_1 \times P_2} \times M_{P_2 \times P_3} \times \cdots \times M_{P_{n-1} \times P_n}$$

Output :- $M_{P_0 \times P_n} = \prod_{i=1}^{n} M_{P_{i-1} \times P_i}$

$M_{p \times q} \times M_{q \times r} \Rightarrow pqr$ multiplications

Example :- $M_{100 \times 10}^{①} \times M_{10 \times 1000}^{②} \times M_{1000 \times 1}^{③} \times M_{1 \times 100}^{④}$

no. of mults $\Rightarrow 100 \times 10 \times 1000 + 1000 \times 1 \times 100 + 100 \times 1000 \times 100$

$①\times②$ $③\times④$ $(①\times②)\times(③\times④)$

$\Rightarrow 111000000$

or $(①(②\times③))④ \Rightarrow 10^3 + 10^4 + 10^4 = 21000$

Q) How many different ways are there to parenthesize a chain of n matrices?

Catalan Numbers $= \frac{1}{n+1} {}^{2n}C_n$

$C_n(i,j) \Rightarrow M_i \times M_j \rightarrow M_{P_{i-1} \times P_i} \times M_{P_i \times P_{i+1}}$

Goal :- To find $C(1,n)$

Easy Step :- $C(i, i+1) = P_{i-1} \times P_i \times P_{i+1}$ || $C(i,i) = 0$

~~Suppose $M_1 \times M_2 \cdots M_{k-1} \times M_k$~~

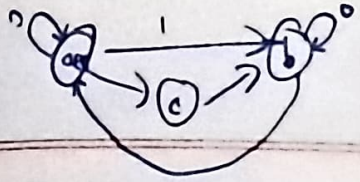
$$C(i,j) = \min_{i \le k < j} \begin{cases} C(i,k) + C(k+1, j) \\ P_i P_k P_j \end{cases}$$

	1	2	3	4
1	0	10^3	$5 \cdot 10^3$	$5 \cdot 10^3$
2		0	10^4	$11 \cdot 10^3$
3			0	10^5
4				0

Goal

$C(1,3) = \min \begin{cases} C(1,2) + C(3,3) + P_0 P_2 P_3 \\ C(1,1) + C(2,3) + P_0 P_1 P_3 \end{cases}$

Similarly



Knapsack :-

$$f(w,i) = \begin{cases} f(w-a_i, i+1) & a_i \le w \\ f(w, i+1) & \text{else} \end{cases}$$

Base Case :-

$$\forall i,\ f(0,i) = 0$$

for i in N :

  for w in [1, ..., x] → This is wrong. use [x, ..., 1] instead

    $dp(w) = dp(w-a_i)$

Shortest Reliable Paths

Input :- Edge wtd graph s, t & k int k

Output :- The shortest path from s to t in G using ≤ k hops/edges

Subproblems :-

$\forall x \in V,\ \forall i \le k$

Let dist(u,i) be the length of the shortest path from s to u, using i hops.

Main Problem :- $\min_{i \le k} dist(t,i)$

Recurrence Relation :-

$$dist(u,i) = \min_{(v,u) \in E} (dist(v, i-1) + w(v,u))$$

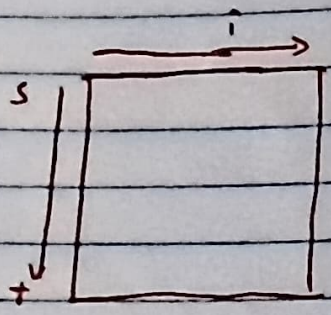
Easiest Subproblems:-

$dist(s,0) = 0$

~~$dist(s,v,0) = \infty$~~

$dist(u,s,0) = \infty$

Table



All pair shortest paths

Input :- G

Output :- $\forall u, v$, length of shortest path from u, v

Subproblems :- Let $dist(u,v,k)$ be the length of the shortest path from u to v using intermediate nodes amongst $[1, k]$

Main Problem :-

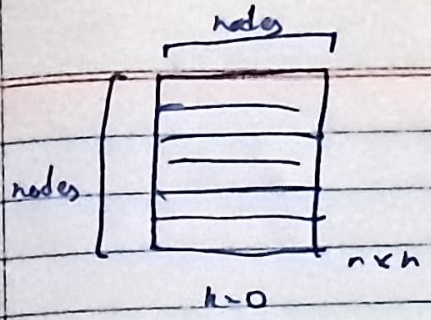
$k = |V|$

Recurrence Relation :-

$$dist(u,v,k) = \min\left(dist(u,v,k-1),\ dist(u,k,k-1) + dist(k,v,k-1)\right)$$

Easiest Subproblem :-

$$dist(u,v,0) = \begin{cases} w(u,v) & \text{if } (u,v) \in E \\ \infty & \text{else} \end{cases}$$



And similarly

Runtime = $O(n^3)$

↓

Same as n dijkstras with fib heaps